Advanced DataBases

Tutorial #1

Exercice 01:

Consider the following schema representing an equipment repair workshop database:

Worker (<u>WorkerNum</u>, WorkerName)

Machine(<u>MachineNum</u>, MachineName, MachineType, RWorkshopNum*)

Repair(<u>RepairNum</u>, MachineNum*, RepairDate)

Worker_Repatation (<u>WorkerNum*, RepairNum*,</u> TimePassed)

RepairWorkshop(<u>RWorkshopNum</u>, WorkshopName)

Express the following queries in the relational Algebra language:

1. Give the worker's names who performed reparations between 15/01/2025 and 30/01/2025.
2. Give the machine's names and types which had been repaired by the worker '*WK001*' in the workshop *"WR004"*.
3. Give the names of the workshops in which the machines have been repaired before 01/01/2025.
4. Give the names of the machines repaired in the workshop in which the machine *'Printer Kyocera M213'* is repaired.
5. Give the workers who have never repaired a machine of type *'IT Equipment'*.
6. Give the names of the workers who repaired all the machines.

Exercice 02:

Express the algebraic queries related to the questions 1, 2, 3 and 4 of the previous exercise by using a tree syntax i.e. the Algebraic tree expression.

Additional exercise:

Let's consider a relational database named PFS with the following schema:

Factory (FN, F_Name, F_City);

Product (PN, P_Name, Color, Weight);

Supplier (SN, S_Name, Status, S_City);

PFS (FN*,PN*,SN*, Quantity);

Where: FN references Factory.FN

PN references Product.PN

SN references Supplier.SN

The database PFS describes what follows:

Factory: a factory is described by a unique number *FN*, its name *F_Name* and the city *F_City* where it is located

Product: a product is described by a unique number *PN*, its name *P_Name*, its *color* and *weight*.

Supplier: a supplier is described by a unique number *SN*, its name *S_Name*, *status* (*client_supplier*, *sub-contractor_supplier*, etc.) and the city *S_City* where he is domiciled

- Answer the following questions by using the relational algebraic language:
    1- Give the factories' numbers, names and cities.
    2- Give the factories' numbers, names and cities of London
    3- Give the suppliers' numbers who supply the product n° 1 for the factory n° 1.
    4- Give the products' names and colors delivered by the supplier n° 1.
    5- Give the suppliers' numbers who supply a red product for the factory n° 1.
    6- Give the products' numbers that are delivered to a factory in London by a supplier in London.

- Express the algebraic queries related to the questions 3, 4, 5 and 6 of the previous exercise by using a tree syntax i.e. the Algebraic tree expression.